

MERCHANT RISK INTELLIGENCE

Quarterly Risk Review - Q4 2024 | NovaPay Sub-Merchant Portfolio

600

TOTAL MERCHANTS

567

ACTIVE (94.5%)

21.08

AVG RISK SCORE

26

RECENT WARNING MERCHANTS

15

MERCHANTS, 3+ RISK SIGNALS*

EXECUTIVE SUMMARY

The NovaPay sub-merchant portfolio (600 merchants across 10 PayFac partners) remains predominantly low-risk: 92.7% of merchants sit in the Low annual-risk band. A small, clearly-defined group requires continued attention - 15 merchants trigger 3 or more of five independent risk-screening signals, and 26 merchants have shown at least one recent (Jul-Dec) risk-deterioration signal. Two merchants, M0055 and M0281, triggered both recent-warning patterns and are the clearest near-term monitoring priorities. Portfolio transaction volume grew through August, dipped in September-October, and recovered by December. Travel carries the largest transaction exposure despite Retail having the most merchants.

Synthetic, portfolio-learning data. Risk indicators are screening signals, not proof of misconduct or causation.

Portfolio Size & Composition

600 merchants across 9 industries, 12 countries, and 10 PayFac partners

600

Total merchants

567

Active (94.5%)

10

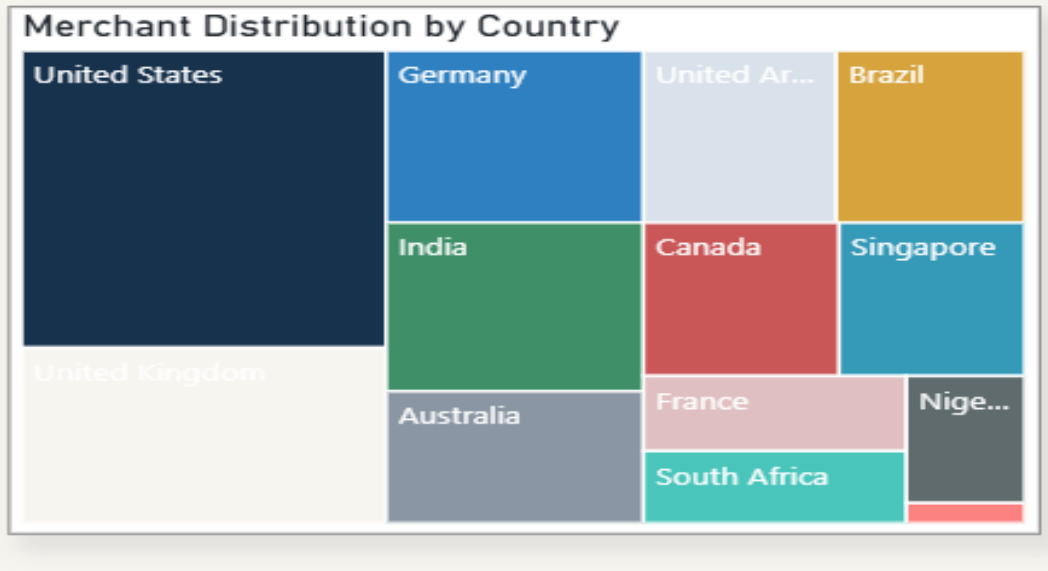
PayFac partners

9 / 12

Industries / Countries

MERCHANT DISTRIBUTION

- Retail is the largest industry (114 merchants, 19%); the top 4- industries hold 57% of the portfolio (moderate concentration).
- PF01 (99) and PF02 (97) manage the largest sub-merchant populations; PF10 the smallest (28).
- United States (137) and United Kingdom (81) are the largest country populations; 3 merchants have missing country data.



Source: Power BI — Portfolio & Concentration

PAYFAC SNAPSHOT

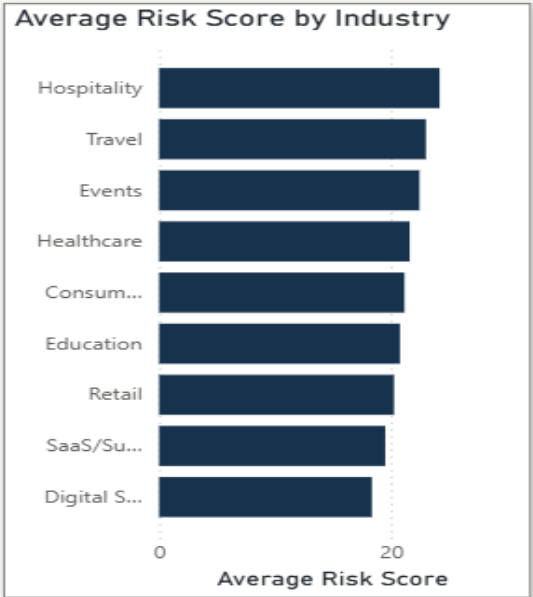
- PF02 generates the highest transaction exposure (\$177.4M) despite PF01 having more merchants.
- PF10 carries the highest average risk score (22.68) across its smaller 28-merchant book.

Industry & Geographic Concentration

Where merchant count, transaction exposure, and average risk diverge



Exposure % vs merchant share %, by industry



Average risk score by industry

Highest avg. risk (industry): Hospitality 24.2, Travel 23.0, Events 22.5

Highest avg. risk (country): Nigeria 25.5, Singapore 23.8, India 22.4

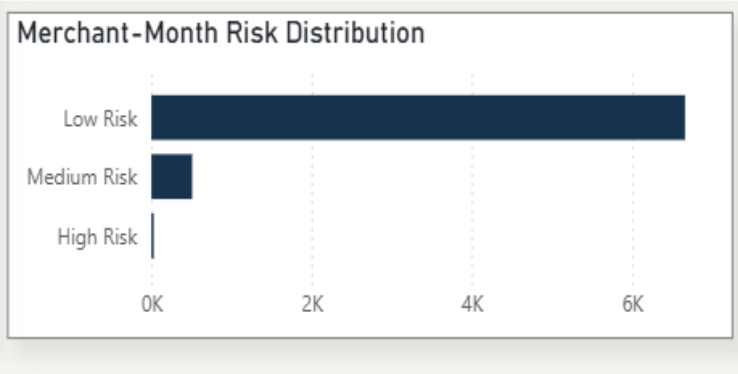
KEY CONCENTRATION FINDING

Travel generates the highest transaction exposure (\$340.3M) even though Retail has the most merchants (114 vs. 74). Merchant count and financial exposure are not concentrated in the same industries, a portfolio review based on merchant counts alone would understate Travel's importance and overstate Retail's.

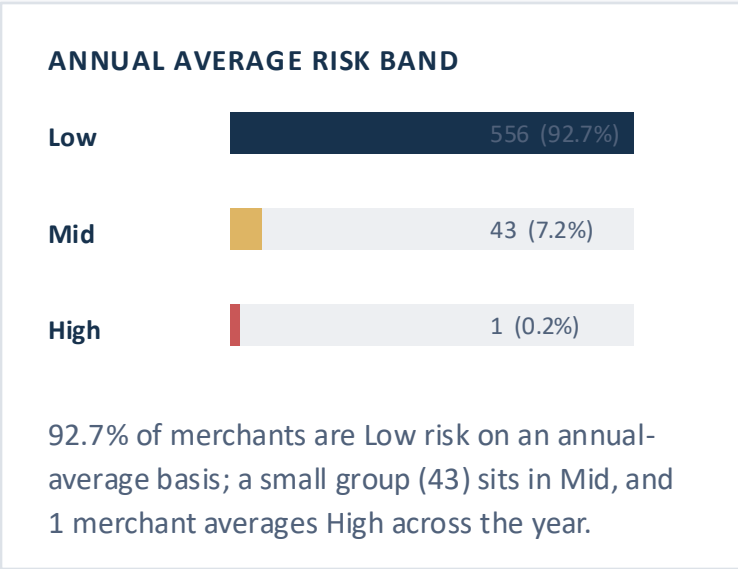
- Industry transaction exposure (top): Travel \$340.3M, Events \$194.7M, Hospitality \$179.8M, Education \$165.1M.
- Industry transaction exposure (bottom): SaaS/Subscriptions \$37.9M, Digital Services \$20.9M.
- Country risk averages are descriptive portfolio statistics from synthetic data, not external risk benchmarks.
- Geographic concentration: US and UK together hold 36% of the merchant base.

How Risk Is Distributed Across the Portfolio

Merchant-month and annual-average risk bands, and the volume/risk relationship



Merchant-month observations (n = 7,200)

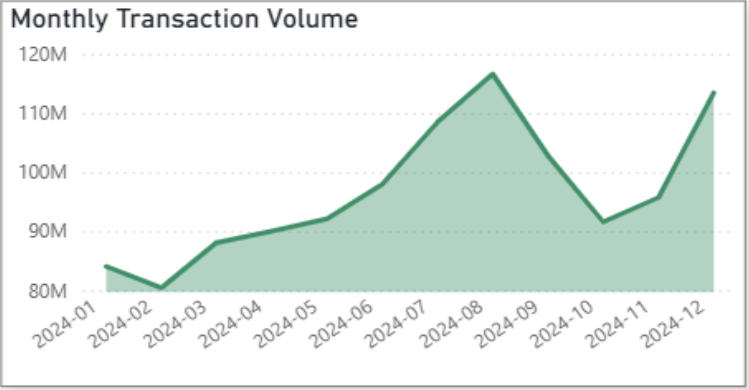


VOLUME ↔ RISK: WEAK RELATIONSHIP, NOT NO RELATIONSHIP

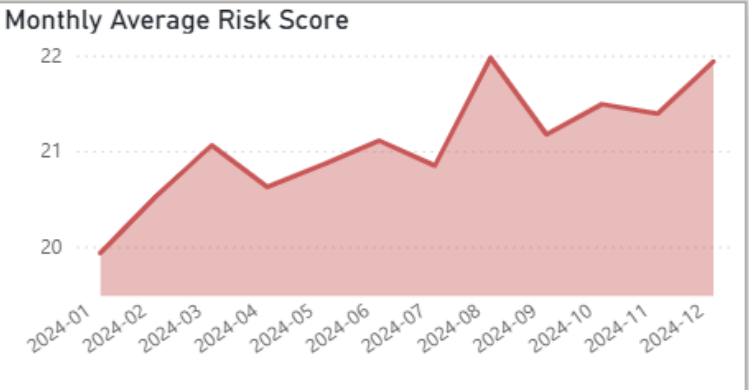
- Average transaction volume and average risk score show almost no linear association across the portfolio (Pearson $r \approx 0.02$) - Transaction volume alone showed little linear association with average risk.
- But the two are not irrelevant to each other: 41 merchants combine above-P75 transaction volume (>\$99,956.53/month) with above-P75 average risk (≥ 22.67) - a smaller, exposure-weighted group worth prioritizing over risk score alone.
- Highest-risk examples in this group include M0350 (avg risk 64.17), M0424 (62.00), and M0280 (53.33).
- Volume bands across the full portfolio: 149 Low-volume, 301 Medium-volume, 150 High-volume merchants (P25/P75 split).

Risk & Volume Trends Through 2024

Portfolio-level movement across the year, and where deterioration concentrated



Monthly transaction volume



Monthly average risk score

FULL-YEAR & RECENT (JUL-DEC) SIGNALS

- Transaction volume rose through August, declined sharply in September-October, then recovered in November-December, with the largest month-over-month gain in December.
- 64 merchants showed risk increases in more than 6 of 11 month-to-month comparisons across the full year, repeated upward movement, not just isolated spikes.
- 20 full-year sudden-deterioration events (≥ 15 -point month-over-month jump) were identified across 19 merchants; August alone accounted for the two largest (M0075, M0496, both +31).
- Restricting to the most recent period (Sep-Dec), 7 sudden-deterioration events occurred, and 13 merchants showed repeated increases in ≥ 5 of the last 6 months (Jul-Dec).
- Combining both recent-window signals: 26 merchants show at least one recent warning sign; M0055 and M0281 are the only two triggering both patterns.

Merchant Risk Screening

Screening candidates from persistent risk, exposure, and multi-signal analysis

Persistently Elevated Merchants

merchant_id	Ave Risk Score	Max Risk Score	Min Risk Score	payfac_id	industry
M0257	65.42	72	58	PF06	Travel
M0350	64.17	70	58	PF05	Travel
M0407	63.50	72	54	PF10	Hospitality
M0424	62.00	71	55	PF03	Hospitality

Persistently elevated merchants — top 4 of 5, by average annual risk score

OTHER SCREENING RESULTS

- 33 merchants exceeded a portfolio P95 chargeback rate (≥1.39%) or fraud rate (≥0.64%), including M0257, M0424, M0407, and M0200.
- 23 merchants carry an adverse-media flag; their average risk score is 29.4, vs. 20.7 for merchants without the flag, an association, not proof of causation.
- Website-risk score shows only a weak positive association with overall merchant risk ($r \approx 0.235$), not a standalone indicator on its own.

MULTI-SIGNAL SCREENING

Five independent signals - elevated average risk, repeated increases, sudden deterioration, elevated chargeback/fraud, and adverse media - combine into a screening count per merchant.

15

merchants trigger 3+ of 5 signals*

- M0055 and M0281 also triggered both recent (Jul–Dec) warning patterns - the clearest continued-monitoring priorities.
- These are investigation priorities based on overlapping indicators, not evidence of problematic activity.

*SQL full-year screening identifies 16; Power BI currently shows 15 pending a definitional reconciliation. See Slide 8.

Data Quality Findings

Issues identified and resolved before portfolio and risk analysis began

Duplicate records

ISSUE

7 exact duplicate merchant-month rows (4 in monthly_transactions, 3 in merchant_risk).

RESOLUTION

Removed via SELECT DISTINCT; validated no duplicates remain.

Inconsistent categories

ISSUE

2 inconsistent 'Retail' industry values (trailing space, capitalization).

RESOLUTION

Standardized to 'Retail'; validated via re-check.

Invalid values

ISSUE

4 invalid negative values (2 transaction_count, 2 transaction_volume).

RESOLUTION

Corrected using ABS(); these were clearly-positive metrics with a sign error.

Missing values (NULLs)

ISSUE

Found in country, delivery_days, reputation_score, company_age_years, ownership_flag.

RESOLUTION

Retained — insufficient evidence to defensibly impute; flagged for source follow-up.

Final validation confirmed no duplicate merchant-month records and no negative transaction values remained. The cleaned data was used for all portfolio and risk analysis in this report.

Recommendations, Methodology & Limitations

PRIORITIZE FOR REVIEW

- M0055 and M0281, the only merchants triggering both recent warning patterns
- Persistently elevated: M0257, M0350, M0407, M0424, M0141
- The 41-merchant high-volume + high-risk-group, weighted by exposure rather than risk score alone

MONITOR & FOLLOW UP

- PF10, highest average risk (22.68) on a small, concentrated 28-merchant book
- 33 merchants exceeding chargeback/fraud P95 thresholds
- 3 merchants with missing country data, source data follow-up recommended

RESOLVE BEFORE NEXT CYCLE

- Reconcile the SQL (16) vs. Power BI (15) multi-signal merchant count, traced to a narrower recent-window definition on two signals in the DAX model
- Confirm treatment of retained NULLs (country, delivery_days, reputation_score, company_age_years, ownership_flag) once source data is available

METHODOLOGY & LIMITATIONS

Built on synthetic, portfolio-learning data for the fictional NovaPay sub-merchant network (600 merchants, 12 months, 2024). The risk score, risk bands, and all screening thresholds are dataset-derived and illustrative, they are not anybody's proprietary methodology. All risk indicators (chargeback/fraud rates, adverse media, website risk, deterioration signals) are screening signals intended to prioritize review, not proof of misconduct or causation. Correlations reported (volume↔risk $r \approx 0.02$; website risk↔overall risk $r \approx 0.235$) are descriptive associations only. A minor definitional gap between the SQL and Power BI multi-signal counts (16 vs. 15) is noted above and pending final reconciliation.